

**IT-302**  
**B.Tech. III Semester**  
**Examination, December 2025**  
**Grading System (GS)**  
**Discrete structure**  
**Time: Three Hour**

**Maximum Marks: 70**

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

- 1 (a) Find the number between 1 to 500 that are not divisible by any of the integers 2 or 3 or 7.
- (b) Prove that  $5^{2n-1}$  is divisible by 24, where  $n$  is any positive integer.
- 2 (a) What is mapping? Explain types of mapping.
- (b) Define relation with example. Explain various types of representation of relation.
- 3 (a) Prove that  $G = \{0, 1, 2, 3, 4, 5, 6\}$  is an abelian group of order 7 with respect to addition modulo 7.
- (b) What is a Ring? Define elementary properties of ring with example.
- 4 (a) Construct truth table of the following and prove:
  - a)  $(p \leftrightarrow p \wedge r) \rightarrow (\sim r \rightarrow \sim p)$  is a tautology
  - b)  $(p \wedge q) \wedge (\sim p) \wedge (\sim q)$  is a contradiction.

(b) Define and explain any four of the following with suitable example.

- i) Bipartite graph
- ii) Hamiltonian path and circuit
- iii) Chromatic number and graph coloring of graph
- iv) Adjacency matrix of graph
- v) Binary search tree

5 (a) Prove that  $F = \{a + b * \sqrt{2} : a, b \text{ rational}\}$  is a field.

(b) Prove or disprove that intersection of two normal subgroups of a group  $G$  is again a normal subgroup of  $G$ . Define subgroup, normal subgroup, Quotient group, with an example for each.

6 (a) Explain the following

- i) Euler Graph
- ii) Isomorphic Graphs
- iii) Minimal spanning tree
- iv) Height of the tree

(b) Define planar Graph. Prove that for any connected planar graph,  $v - e + r = 2$  where  $v, e, r$  is the number of vertices, edges, and region of the graph respectively.

7 (a) Solve the following recurrence relation :

$$a_r - 7a_{r-1} + 10a_{r-2} = 3^r \text{ Given that } a_0 = 0, a_1 = 1.$$

- (b) Solve the following recurrence relation using generating function

$$G(K) - 7G(K-1) + 10G(K-2) = 8K + 6$$

- 8 (a) Discuss in brief any two of the following:

- i) Partial ordering relation
- ii) Poset
- iii) Disjunctive normal form
- iv) Pigeonhole principle

- (b) Define the lattice. Let  $(L, \leq, \vee, \wedge)$  be a lattice and  $a, b, c, d \in L$  such that  $a \leq b$  and  $c \leq d$ . Show that  $a \vee c \leq b \vee d$  and  $a \wedge c \leq b \wedge d$ .